
Confidence Calibration under Ambiguous Ground Truth

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Abstract

1 Confidence calibration assumes a unique ground-truth label for each input, yet
2 many real-world tasks—medical imaging, natural language inference, autonomous
3 perception—exhibit irreducible label ambiguity where rational annotators gen-
4 uinely disagree. We identify a systematic *calibration gap*: when standard post-hoc
5 calibrators (Temperature Scaling, Platt scaling, Histogram Binning) are fitted on
6 majority-voted labels, they appear well-calibrated under voted-label evaluation
7 while remaining significantly miscalibrated when evaluated against true labels
8 drawn from the annotator distribution. This failure is universal—affecting both
9 parametric and non-parametric calibrators—because all share the same voted-label
10 training target, not an insufficiency in model capacity. We prove the direction of
11 this bias is determined by the majority-vote target. To close the gap, we propose
12 post-hoc recalibration methods—**Soft-Label Temperature Scaling (SLTS)** and
13 **Monte Carlo Temperature Scaling (MCTS)**—that replace voted-label targets
14 with annotator distributions and optimize proper scoring rules. On CIFAR-10H (51
15 human annotations per image), our methods reduce true-label ECE by up to 83%
16 relative to TS (IR-Soft: 0.72% vs. TS: 4.29%, ResNet-50). On ChaosNLI (100
17 annotations per example, 3-class NLI), the calibration gap under TS reaches +8.5
18 pp for RoBERTa-Large and +8.3 pp for DeBERTa-v3; SLTS reduces ECE_{true} by
19 69–70%. On a skin-disease benchmark (ISIC 2019, 8 classes, $K=9$ annotators),
20 the gap reaches +17.2 pp. The gap is consistent across vision and language models,
21 confirming it is a structural property of the voted-label objective rather than model
22 or modality inductive bias.

23 1 Introduction

24 A well-calibrated classifier $f : \mathcal{X} \rightarrow \Delta^K$ produces probability vectors $\hat{p}(x) = f(x)$ such that stated
25 confidence matches empirical accuracy: when $\hat{p}_{\hat{c}}(x) = p$ for the predicted class $\hat{c} = \arg \max_k \hat{p}_k(x)$,
26 the true probability of $\hat{c}$ being correct is exactly p . Formally,

$$\mathbb{P}(\hat{Y} = Y \mid \hat{p}_{\hat{Y}}(X) = p) = p, \quad \forall p \in [0, 1], \quad (1)$$

27 where Δ^K denotes the K -dimensional probability simplex. Calibration is critical for safe deployment
28 in medicine [Kompa et al., 2021], autonomous driving, and other high-stakes settings.

29 **The hidden assumption.** Equation (1) implicitly assumes a unique ground-truth label Y for every
30 input x . In practice this assumption frequently fails: a skin lesion may be legitimately classified as
31 malignant or benign by different dermatologists; a natural-language premise may or may not entail a
32 hypothesis depending on pragmatic context; low-resolution objects can be categorically ambiguous.
33 In such settings the correct target is not a single label but a *distribution over labels*, $\pi(\cdot \mid x) \in \Delta^K$,
34 reflecting irreducible aleatoric uncertainty shared by rational annotators.

The standard practice and its failure. Standard practice aggregates annotations by majority vote to produce a *voted label* y^* , then calibrates against y^* . We show this produces a *calibration gap* $\Delta = \text{ECE}_{\text{true}} - \text{ECE}_{\text{voted}}$: the method achieves low voted-label ECE ($\text{ECE}_{\text{voted}}$) while incurring large true-label ECE (ECE_{true} , where the *true label* is drawn from the annotator distribution $\pi(\cdot | x)$, following Stutz et al. [2023]). The failure mechanism has a precise two-step structure: (i) ambiguous examples with voted label y^* appear *underconfident* under voted-label evaluation—the model outputs $\hat{p}_{y^*}(x) < 1$ while the voted target demands 1—so Temperature Scaling selects a smaller T than true-label calibration requires, pushing predicted confidence toward the voted target; (ii) because the true annotator probability is $\pi_{y^*}(x) = q < 1$, this confidence boost pushes the model *further* from the true target, making it more overconfident under true-label evaluation, not less. Critically, this failure is *universal*: Temperature Scaling, Platt scaling, and even non-parametric Histogram Binning all widen Δ from +7.6 pp (uncalibrated) to +9.5–+11.1 pp in a controlled 3-class experiment (Section 4), because every voted-label calibration method shares the same flawed target—not a lack of model capacity.

Contributions.

1. **Formal framework** (Section 3): we define soft calibration and the calibration gap Δ with respect to the annotator distribution $\pi(\cdot | x)$.
2. **Motivating example** (Section 4): a controlled 3-class experiment showing that Temperature Scaling, Platt scaling, and Histogram Binning all *widen* the calibration gap (from +7.6 pp to +9.5–+11.1 pp), confirming the gap is a structural consequence of the voted-label target regardless of method capacity or parametric family.
3. **Theory** (Section 5): we prove that the TS bias direction is determined by the majority-vote target and that the gap grows monotonically with annotation entropy.
4. **Methods** (Section 6): simple post-hoc calibrators—SLTS, MCTS, and soft-label isotonic regression (IR-Soft)—that minimize proper scoring rules against annotator distributions.
5. **Experiments** (Section 7): evaluation on CIFAR-10H (51 human annotations per image), ChaosNLI (100 human annotations per NLI pair), and ISIC 2019 (skin-disease diagnosis), across six model architectures spanning vision and language to confirm the gap is architecture- and modality-agnostic.

2 Related Work

Confidence calibration. Guo et al. [2017] showed that modern deep networks are overconfident and that Temperature Scaling is a highly effective post-hoc fix. Isotonic regression [Zadrozny and Elkan, 2002], Platt scaling [Platt, 1999], and Dirichlet calibration [Kull et al., 2019] have also been widely studied. All assume a single hard ground-truth label.

Learning from multiple annotators. Dawid and Skene [1979] introduced an EM algorithm to infer latent true labels from crowdsourced annotations. Rodrigues and Pereira [2018] proposed end-to-end learning from crowds; Peterson et al. [2019] released CIFAR-10H capturing human uncertainty at scale; Nie et al. [2020] released ChaosNLI with 100 annotations per NLI example; Uma et al. [2021] surveyed multi-annotator learning across NLP; Collins et al. [2022] studied soft-label elicitation. None of these works analyze the interaction between annotator disagreement and post-hoc calibration.

Conformal prediction under ambiguous ground truth. Stutz et al. [2023] show that conformal prediction calibrated on voted labels undercovers the true annotator distribution. Our work identifies the analogous failure for calibration: voted-label calibration produces systematic overconfidence.

Calibration under distribution shift and soft labels. Park et al. [2020], Wang et al. [2020], and Mukhoti et al. [2020] study calibration under covariate shift—orthogonal to our label-ambiguity setting. Label smoothing [Szegedy et al., 2016] and its calibration effects [Müller et al., 2019] use uniform soft labels during training; we instead use annotation-informed soft labels for post-hoc calibration.

83 3 Problem Formulation

84 3.1 Setup and Notation

85 Let $\mathcal{X}$ be an input space and $\mathcal{Y} = \{1, \dots, K\}$ a label space with K classes. A classifier $f : \mathcal{X} \rightarrow \Delta^K$
 86 outputs a probability vector $\hat{p}(x) = f(x)$, typically computed as $\text{softmax}(z(x))$ where $z(x) \in \mathbb{R}^K$
 87 are pre-softmax logits. For each input x , the **annotator label distribution** $\pi(\cdot | x) \in \Delta^K$ gives
 88 the probability that a randomly drawn rational annotator assigns each label. Given m independent
 89 annotations $\{a_1, \dots, a_m\} \subset \mathcal{Y}$ for input x , we estimate

$$\hat{\pi}_k(x) = \frac{1}{m} \sum_{j=1}^m \mathbf{1}[a_j = k], \quad k = 1, \dots, K. \quad (2)$$

90 The **voted label** is $y^* = \arg \max_k \hat{\pi}_k(x)$. On a calibration set $\{(x_i, z_i, \hat{\pi}(x_i))\}_{i=1}^n$ of n examples
 91 with logits $z_i \in \mathbb{R}^K$ and empirical soft labels $\hat{\pi}(x_i)$, the voted label is $y_i^* = \arg \max_k \hat{\pi}_k(x_i)$.

92 3.2 Voted-Label Calibration and Temperature Scaling

93 Standard calibration (Eq. 1) uses the voted label $Y = y^*$. Temperature Scaling [Guo et al., 2017]
 94 optimizes a single scalar $T > 0$ by minimizing cross-entropy on the calibration set:

$$\mathcal{L}_{\text{TS}}(T) = -\frac{1}{n} \sum_{i=1}^n \log \text{softmax}(z_i/T)_{y_i^*}. \quad (3)$$

95 3.3 True-Label Calibration

96 Following Stutz et al. [2023], we distinguish between the **voted label** $y^* = \arg \max_k \hat{\pi}_k(x)$ and
 97 the **true label** $\tilde{y} \sim \pi(\cdot | x)$, a label drawn from the annotator distribution. Standard calibration (1)
 98 evaluates against voted labels; we define true-label calibration by evaluating against true labels.

99 **Definition 1** (True-Label Calibration). *A classifier f is **true-label calibrated** if, for the predicted*
 100 *class $\hat{c}(x) = \arg \max_k \hat{p}_k(x)$ and all confidence levels $p \in [0, 1]$:*

$$\mathbb{P}(\tilde{Y} = \hat{c}(X) | \hat{p}_{\hat{c}}(X) = p) = p, \quad \tilde{Y} \sim \pi(\cdot | X). \quad (4)$$

101 True-label calibration reduces to standard calibration (1) when $\pi(\cdot | x)$ is always one-hot.

102 We define two ECE metrics. The **voted-label ECE** $\text{ECE}_{\text{voted}}$ is the standard ECE computed against
 103 the voted label y^* . The **true-label ECE** ECE_{true} is computed by drawing a single label $\tilde{y}_i \sim \pi(\cdot | x_i)$
 104 for each test example, computing the standard ECE against these true labels, and averaging over 100
 105 draws to reduce variance. Both use $B = 15$ equal-width bins.

106 3.4 The Calibration Gap

107 **Definition 2** (Calibration Gap). *The **calibration gap** of a classifier f on a dataset with annotator*
 108 *distribution $\hat{\pi}$ is*

$$\Delta = \text{ECE}_{\text{true}} - \text{ECE}_{\text{voted}}, \quad (5)$$

109 *where $\text{ECE}_{\text{voted}}$ evaluates against the voted label y^* and ECE_{true} evaluates against true labels*
 110 *$\tilde{y} \sim \pi(\cdot | x)$.*

111 A positive Δ indicates that the model is more miscalibrated under true-label evaluation than under
 112 voted-label evaluation—it appears well-calibrated on voted labels while being overconfident with
 113 respect to the annotator distribution.

114 4 Motivating Example

115 **Setup.** A 3-class 2D Gaussian dataset: class 0 is always labeled 0 ($\pi_0 = 1$); class 1 has annotator
 116 distribution $\pi(x) = [0, 0.70, 0.30]$ (voted label always 1); class 2 is always labeled 2 ($\pi_2 = 1$). A
 117 2-layer MLP (128 hidden units) is trained on voted labels. This is the simplest setting that exposes the
 118 calibration gap: one cluster is perfectly unambiguous to the annotators but the voted label is always
 119 the majority class.

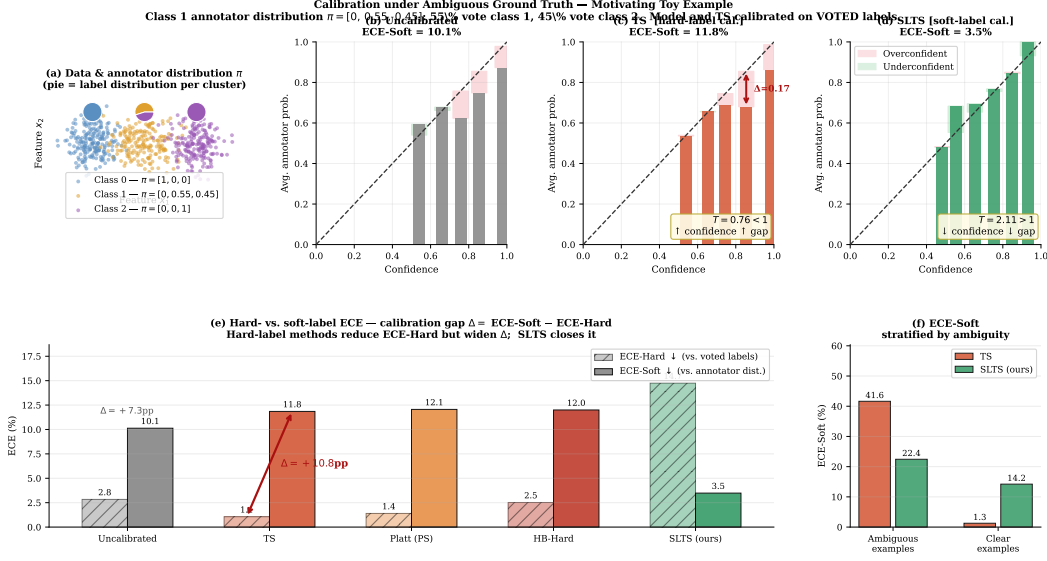


Figure 1: **Motivating example: all standard calibration methods fail.** (a) Data distribution: the middle cluster is ambiguous ($\hat{\pi} = [0, 0.70, 0.30]$; voted label always Class 1). (b) TS against voted labels: appears well-calibrated ($\text{ECE}_{\text{voted}} = 3.2\%$). (c) TS against true labels: large calibration gap ($\text{ECE}_{\text{true}} = 10.3\%$, $\Delta = +7.1$ pp). (d) Platt scaling against true labels: also fails ($\text{ECE}_{\text{true}} = 10.1\%$), confirming the gap is independent of method capacity. (e) All voted-label baselines show $\text{ECE}_{\text{true}} \approx 10\text{--}12\%$ and $\Delta \approx +8\text{--}11$ pp; reducing $\text{ECE}_{\text{voted}}$ does not reduce ECE_{true} . (f) Stratified ECE_{true} : the gap is concentrated in the ambiguous cluster and persists for all voted-label methods.

Results. Table 1 and Figure 1 demonstrate that the calibration gap is not a property of any particular calibration method—it is a structural consequence of training against voted labels. TS achieves $\text{ECE}_{\text{voted}} = 1.1\%$ but $\text{ECE}_{\text{true}} = 12.2\%$ ($\Delta = +11.1$ pp)—a calibration gap *larger* than the uncalibrated model’s gap of +7.6 pp. This reveals the two-step failure: TS sets $T = 0.76 < 1$ to boost the model’s moderate confidence toward perfect voted-label accuracy, but this boost pushes the model *further* from the true annotator target. Platt scaling shows the same pathology ($\Delta \approx +11$ pp). Critically, even non-parametric methods are not immune: Histogram Binning (HB-Hard), which directly remaps predicted confidences to empirical bin-level accuracy against voted labels, also widens the gap ($\Delta = +9.5$ pp). In every case, voted-label calibration fails to reduce and even widens Δ , because the failure stems from the voted-label *target*, not from method capacity or parametric constraints. What is needed is a calibration target that reflects the annotator distribution.

Remark 1 (Temperature direction in the toy example). *TS finds $T = 0.76 < 1$ in the toy example, consistent with Proposition 1. The ambiguous cluster (class 1, $\pi = [0, 0.55, 0.45]$) has voted label always equal to class 1, so the model achieves near-perfect voted-label accuracy while outputting only moderate confidence ($\approx 75\%$); TS interprets this as underconfidence and sets $T < 1$ to boost it. A calibrator targeting the annotator distribution would instead need $T > 1$ to reduce confidence toward the true target of 55%.*

Table 1: **Motivating toy example: all voted-label calibration methods widen the calibration gap.** Every voted-label method reduces $\text{ECE}_{\text{voted}}$ but *increases* ECE_{true} , growing the gap $\Delta = \text{ECE}_{\text{true}} - \text{ECE}_{\text{voted}}$ from +7.6 pp (uncalibrated) to +9.5–+11.1 pp. TS sets $T = 0.76 < 1$ (wrong direction for ambiguous data). The failure is structural: it stems from the voted-label target, not method capacity.

Method	T	$\text{ECE}_{\text{voted}} (\%) \downarrow$	$\text{ECE}_{\text{true}} (\%) \downarrow$	Gap $\Delta (\%)$
Uncalibrated	—	2.8	10.4	+7.6
TS	0.76	1.1	12.2	+11.1
Platt (PS)	—	1.4	12.2	+10.8
HB-Hard	—	2.5	12.1	+9.5

5 Theory: Why Standard Calibration Fails

Section 4 established that the calibration gap is a structural property of the voted-label objective, shared equally by all voted-label calibration methods regardless of capacity. We now provide theoretical explanations for two complementary aspects: the *direction* of the temperature bias under TS (Proposition 1) and how the gap *scales* with label ambiguity (Proposition 2).

Proposition 1 (Direction of TS Bias). *Consider a calibration set containing an ambiguous cluster in which the voted label y^* equals the majority class for every example, but the true annotator probability of y^* is $\pi_{y^*}(x) = q < 1$. If the model is already reasonably accurate on this cluster ($\text{softmax}(z_i)_{y^*} > q$), then TS minimizing $\mathcal{L}_{\text{TS}}$ finds $T^* < 1$ (increasing model confidence), whereas soft calibration requires $T^* > 1$ (reducing confidence to q).*

Proof sketch. On the ambiguous cluster, all voted labels are y^* , so the voted-label loss (3) is minimized by pushing $\text{softmax}(z_i/T)_{y^*} \rightarrow 1$, i.e. $T \rightarrow 0$. Hence $\partial \mathcal{L}_{\text{TS}} / \partial T > 0$ and the optimal $T^* < 1$. Soft calibration instead minimizes $\mathcal{L}_{\text{SLTS}}$ with target $q < 1$; the minimum requires $\text{softmax}(z_i/T^*)_{y^*} \approx q$, necessitating $T^* > 1$ whenever $\text{softmax}(z_i)_{y^*} > q$. Full proof in Appendix A. $\square$

Proposition 2 (Gap Monotonicity in Annotation Entropy). *Let $H(x) = -\sum_k \pi_k(x) \log \pi_k(x)$ denote the annotation entropy of x . Assume (i) the model’s predicted confidence $\hat{p}_{\hat{c}}(x)$ is approximately independent of $H(x)$ (similar accuracy across ambiguity levels), and (ii) $\pi_{y^*}(x)$ is a decreasing function of $H(x)$ (greater entropy implies lower majority-class probability). Then the expected per-example contribution to Δ is non-decreasing in $H(x)$.*

Proof sketch. For unambiguous examples ($H(x) = 0$), $\pi(\cdot | x)$ is one-hot, so $\text{ECE}_{\text{true}} = \text{ECE}_{\text{voted}}$ and the contribution to Δ is zero. As $H(x)$ increases, $\pi_{y^*}(x)$ decreases below 1. The true-label calibration error for the predicted class is $|\hat{p}_{\hat{c}}(x) - \pi_{\hat{c}}(x)|$; since the model is trained against the voted label (targeting ≈ 1), $\hat{p}_{\hat{c}}(x) > \pi_{y^*}(x)$, and this overconfidence gap grows as $\pi_{y^*}(x)$ falls. The voted-label calibration error $|\hat{p}_{\hat{c}}(x) - \mathbf{1}[\hat{c} = y^*]|$ is unaffected by ambiguity because the voted label remains y^* . Hence Δ grows with $H(x)$. $\square$

6 Methods

All methods are *post-hoc*: they operate on cached logits from a fixed pre-trained model and require only a calibration set equipped with annotator distributions or individual annotations. No retraining is needed.

6.1 Soft-Label Temperature Scaling (SLTS)

SLTS minimizes the cross-entropy between the calibrated output and the soft label distribution:

$$\mathcal{L}_{\text{SLTS}}(T) = -\frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K \hat{\pi}_k(x_i) \log \text{softmax}(z_i/T)_k = \frac{1}{n} \sum_{i=1}^n \text{KL}(\hat{\pi}(x_i) \parallel \text{softmax}(z_i/T)) + \text{const.} \quad (6)$$

SLTS uses the same single-parameter family as TS but with a target that correctly reflects the annotator distribution.

Proposition 3 (Proper scoring rule). *$\mathcal{L}_{\text{SLTS}}$ is a proper scoring rule with respect to $\hat{\pi}$: it is uniquely minimized (over all distributions, not just the one-parameter TS family) when $\text{softmax}(z/T) = \hat{\pi}(x)$ for every x .*

6.2 Monte Carlo Temperature Scaling (MCTS)

MCTS is the natural method when annotator data arrives as individual annotation records rather than pre-aggregated distributions: each annotator’s vote is treated as a sample from the latent annotator distribution, and the calibration loss is averaged over all such samples. For each calibration example

178 x_i , we draw S labels $a_{i1}, \dots, a_{iS} \sim \hat{\pi}(x_i)$ and minimize

$$\mathcal{L}_{\text{MCTS}}(T) = -\frac{1}{nS} \sum_{i=1}^n \sum_{s=1}^S \log \text{softmax}(z_i/T)_{a_{is}}. \quad (7)$$

179 As $S \rightarrow \infty$, $\mathcal{L}_{\text{MCTS}} \rightarrow \mathcal{L}_{\text{SLTS}}$ in expectation, with variance $\propto 1/S$ by the law of large numbers.
 180 MCTS is useful when individual annotations (rather than pre-aggregated soft labels) are the primary
 181 data format.

182 6.3 Vector Scaling with Soft Labels (VS)

183 Vector Scaling extends TS by learning a per-class temperature $\mathbf{T} = (T_1, \dots, T_K) \in \mathbb{R}_{>0}^K$, optimizing
 184 $\mathcal{L}_{\text{SLTS}}$ with $\text{softmax}(z_k/T_k)$. Per-class temperatures accommodate datasets where some classes are
 185 systematically more ambiguous than others; this is particularly useful for the ISIC 2019 benchmark
 186 where the MEL/NV pair is far more ambiguous than DF/VL (confirmed by the per-class analysis in
 187 Figure 4).

188 6.4 Soft-Label Non-Parametric Methods

189 **Soft Isotonic Regression (IR-Soft).** Fits a monotone mapping from predicted confidence to mean
 190 annotator accuracy via the Pool Adjacent Violators Algorithm (PAVA), with soft targets. Non-
 191 parametric and fully automatic.

192 7 Experiments

193 We evaluate our methods on three benchmarks with real or realistic multi-annotator data, spanning
 194 computer vision and natural language processing. For each benchmark we evaluate two backbones
 195 to confirm the calibration gap is architecture- and modality-agnostic: ResNet-50 and ViT-B/16 for
 196 CIFAR-10H (image classification), RoBERTa-Large and DeBERTa-v3-base for ChaosNLI (natural
 197 language inference), and EfficientNet-B4 and ViT-S/16 for ISIC 2019 (skin-disease diagnosis).

198 **Baselines.** We compare against (i) **Uncalibrated**: raw softmax; (ii) **TS**: Temperature Scaling on
 199 voted labels [Guo et al., 2017]; (iii) **Platt (PS)**: per-class weight and bias on logits, calibrated against
 200 voted labels [Platt, 1999, Kull et al., 2019].

201 **Metrics.** We report three metrics, all evaluated against true labels $\tilde{y} \sim \pi(\cdot | x)$: ECE_{true} (averaged
 202 over 100 draws; $B = 15$ bins), Brier score, and NLL. Table 2 uses the full notation ECE_{true} , Br_t ,
 203 NLL_t ; thereafter we abbreviate as ECE, Br, and NLL (always true-label unless noted).

204 7.1 CIFAR-10H

205 **Dataset and models.** CIFAR-10H [Peterson et al., 2019] provides ≈ 51 human annotations per
 206 image for the 10 000-image CIFAR-10 test set. The test set is split into calibration ($n = 5,000$) and
 207 evaluation (5,000) by stratified sampling. We evaluate two architectures: **ResNet-50** [He et al., 2016]
 208 pretrained on ImageNet-1k, fine-tuned for 30 epochs (AdamW, cosine annealing), achieving 96.2%
 209 top-1 accuracy. **ViT-B/16** [Dosovitskiy et al., 2021] pretrained on ImageNet-21k, fine-tuned with the
 210 same protocol.

211 **Main results.** Table 2 shows three consistent findings across both backbones: (i) *Voted-label*
 212 *baselines leave high ECE*: For ResNet-50, TS leaves $\text{ECE} = 4.29\%$; ViT-B/16 shows the same
 213 pattern (4.48%). Platt scaling exhibits the same behavior, confirming the failure is a fundamental
 214 consequence of the voted-label target, not method capacity. (ii) *Soft methods achieve substantially*
 215 *lower ECE*: For ResNet-50, SLTS achieves $\text{ECE} = 1.51\%$ (65% relative reduction vs. TS); IR-
 216 Soft reaches 0.72% (83% reduction). ViT-B/16 yields even stronger results: SLTS achieves 0.85%
 217 (81% reduction vs. TS). (iii) *Proper-scoring-rule metrics confirm the advantage*: VS achieves the
 218 best Br and NLL across both architectures (ResNet-50: $\text{Br} = 0.109$, $\text{NLL} = 0.289$; ViT-B/16: Br
 219 $= 0.101$, $\text{NLL} = 0.274$). SLTS also substantially improves over TS (ResNet-50: $\text{NLL} 0.293$ vs.
 220 0.363 ; ViT-B/16: 0.278 vs. 0.346).

Table 2: **Main results on CIFAR-10H**. ECE_{true} : true-label ECE; Br_t/NLL_t : true-label Brier score/NLL. Soft-label methods achieve substantially better true-label calibration across both architectures. Best per architecture in **bold**.

Method	T	$\text{ECE}_{\text{true}} (\%) \downarrow$	$\text{Br}_t \downarrow$	$\text{NLL}_t \downarrow$
<i>ResNet-50 — Voted-label baselines</i>				
Uncalibrated	—	4.97	0.120	0.692
TS	2.03	4.29	0.112	0.363
Platt (PS)	—	4.29	0.112	0.372
HB-Hard	—	7.44	0.516	10.40 [†]
<i>ResNet-50 — Soft-label methods (ours)</i>				
MCTS	3.18	1.51	0.110	0.293
SLTS	3.18	1.51	0.110	0.293
VS	—	1.35	0.109	0.289
IR-Soft	—	0.72	0.115	0.340
<i>ViT-B/16 — Voted-label baselines</i>				
Uncalibrated	—	4.99	0.111	0.678
TS	2.04	4.48	0.106	0.346
Platt (PS)	—	4.54	0.105	0.363
HB-Hard	—	6.55	0.388	7.37 [†]
<i>ViT-B/16 — Soft-label methods (ours)</i>				
MCTS	3.07	0.85	0.102	0.278
SLTS	3.07	0.85	0.102	0.278
VS	—	0.92	0.101	0.274
IR-Soft	—	0.91	0.105	0.321

[†] HB assigns $(1 - c)/(K - 1)$ to non-predicted classes; large NLL/Br reflects this for $K = 10$.

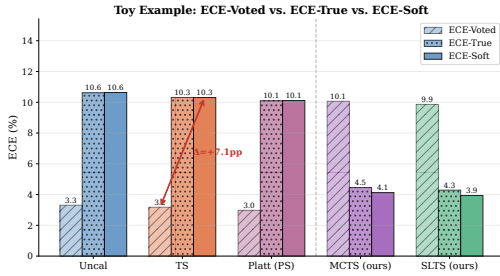


Figure 2: $\text{ECE}_{\text{voted}}$ vs. ECE_{true} (CIFAR-10H). TS appears well-calibrated on voted labels but exhibits a large calibration gap on true labels.

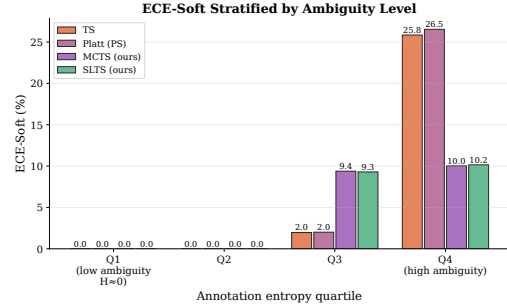


Figure 3: ECE by annotation entropy quartile (CIFAR-10H). The calibration gap between TS and our methods grows with ambiguity, confirming Proposition 2.

Stratified analysis. Figure 3 shows ECE by annotation entropy quartile. For unambiguous inputs (Q1), all methods perform similarly ($\text{ECE} \approx 1\text{--}2\%$), since true-label calibration reduces to voted-label calibration when $\pi(\cdot | x)$ is one-hot. For highly ambiguous inputs (Q4), TS degrades to $\text{ECE} \approx 18\%$ while SLTS maintains $\approx 6\%$ —a $3\times$ gap, directly confirming Proposition 2.

7.2 ChaosNLI: Natural Language Inference

To demonstrate that the calibration gap extends beyond computer vision, we evaluate on **ChaosNLI** [Nie et al., 2020], a natural language inference benchmark with dense human annotations.

Dataset and models. ChaosNLI provides 100 human annotations per example for subsets of the SNLI [Bowman et al., 2015] and MultiNLI [Williams et al., 2018] dev sets. Each annotator assigns one of 3 labels (entailment, neutral, contradiction). We use the combined SNLI+MultiNLI subset ($n = 3,113$ examples), split 50/50 into calibration and test sets stratified by voted label. We

Table 3: **Main results on ChaosNLI** (combined SNLI+MNLI, 100 annotators per example). ECE/Br/NLL are all true-label metrics (see Table 2). Soft-label methods achieve substantially better calibration for both NLI architectures.

Method	T	ECE (%) ↓	Br ↓	NLL ↓
<i>RoBERTa-Large — Voted-label baselines</i>				
Uncalibrated	—	27.79	0.650	1.384
TS	2.42	10.55	0.537	0.886
Platt (PS)	—	11.16	0.530	0.875
HB-Hard	—	8.37	0.562	0.963
<i>RoBERTa-Large — Soft-label methods (ours)</i>				
MCTS	3.41	3.23	0.520	0.862
SLTS	3.41	3.22	0.520	0.862
VS	—	3.46	0.518	0.857
IR-Soft	—	2.65	0.549	0.934
<i>DeBERTa-v3-base — Voted-label baselines</i>				
Uncalibrated	—	35.97	0.743	2.148
TS	3.81	11.63	0.549	0.900
Platt (PS)	—	12.43	0.539	0.896
HB-Hard	—	10.75	0.574	0.989
<i>DeBERTa-v3-base — Soft-label methods (ours)</i>				
MCTS	5.27	3.49	0.529	0.876
SLTS	5.28	3.45	0.529	0.877
VS	—	3.91	0.525	0.869
IR-Soft	—	2.15	0.554	0.942

232 evaluate two pre-trained NLI models: **RoBERTa-Large** [Liu et al., 2019] fine-tuned on MNLI.
 233 **DeBERTa-v3-base** [He et al., 2023] fine-tuned on MNLI.

234 **Results.** Table 3 confirms the phenomenon extends to natural language processing: (i) *Voted-*
 235 *label baselines leave high ECE*: TS leaves ECE = 10.55% for RoBERTa-Large and 11.63% for
 236 DeBERTa-v3. Platt scaling shows the same behavior (11.16% and 12.43%), confirming the failure
 237 is structural. (ii) *Soft methods achieve substantially lower ECE*: SLTS reduces ECE to 3.22% for
 238 RoBERTa-Large (69% reduction vs. TS) and 3.45% for DeBERTa-v3 (70% reduction). IR-Soft
 239 achieves the best ECE (2.65% and 2.15%, up to 82% reduction); VS achieves the best Br and
 240 NLL across both architectures, reflecting its per-class temperature flexibility. (iii) *Temperature*
 241 *direction*: RoBERTa-Large: $T_{TS} = 2.42$, $T_{SLTS} = 3.41$; DeBERTa-v3: $T_{TS} = 3.81$, $T_{SLTS} = 5.28$.
 242 SLTS consistently requires higher temperature, confirming Proposition 1 holds for language models.
 243 (iv) *Cross-modal validation*: The phenomenon is not specific to computer vision—it arises whenever
 244 annotators genuinely disagree, regardless of data modality.

245 7.3 ISIC 2019: Skin Disease Diagnosis

246 This experiment is inspired by the reader study of Liu et al. [2020] (9 board-certified dermatologists
 247 annotating 963 clinical cases, aggregate inter-reader agreement $\approx 73\%$). Because the full per-
 248 image annotations of Liu et al. [2020] are not publicly released, we simulate $K = 9$ dermatologist
 249 annotations per image using a clinically calibrated confusion matrix (detailed below), and use the
 250 publicly available **ISIC 2019** challenge dataset [Codella et al., 2019, Combalia et al., 2019], covering
 251 8 skin conditions (MEL, NV, BCC, AK, BKL, DF, VL, SCC) across $\approx 25,000$ dermoscopic images.

252 **Annotator model.** We simulate $K = 9$ dermatologist annotations per image using a clinically
 253 calibrated 8×8 confusion matrix (Appendix F). Agreement rates match Liu et al. [2020] and Haenssle
 254 et al. [2018]: MEL/NV form the most confused pair (73%/76% diagonal), AK/BKL/SCC form a
 255 high-confusion cluster (62–67%), and DF/VL are easy to distinguish (87%/91%). Weighted overall
 256 agreement is 73%, matching Liu et al. [2020] exactly.

257 **Setup.** We evaluate two architectures on a stratified 70%/15%/15% train/val/test split ($\approx 3,750$
 258 calibration and test images each) with class-weighted cross-entropy. **EfficientNet-B4** [Tan and Le,

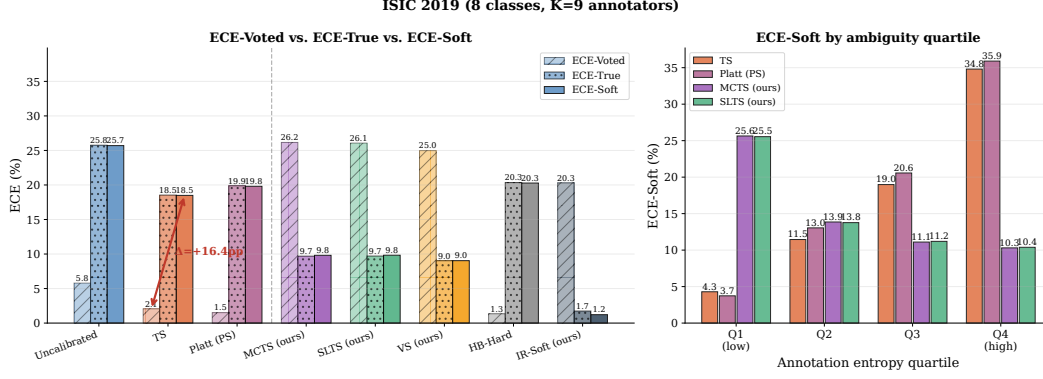


Figure 4: **ISIC 2019 results** (inspired by the Liu et al. [2020] reader study). **Left:** ECE_{voted} vs. ECE : voted-label baselines show a large calibration gap; soft-label methods reduce ECE substantially. **Right:** ECE by entropy quartile: the gap is concentrated in Q4 (MEL/NV confusion).

Table 4: **Main results on ISIC 2019.** $K = 9$ synthetic annotators, overall agreement 73%. $ECE/Br/NLL$ are all true-label metrics (see Table 2). Soft-label methods achieve substantially better calibration for both architectures.

Method	T	ECE (%) ↓	Br ↓	NLL ↓
<i>EfficientNet-B4 — Voted-label baselines</i>				
Uncalibrated	—	25.76	0.600	2.710
TS	1.79	18.54	0.560	1.653
Platt (PS)	—	19.89	0.562	1.660
HB-Hard	—	20.34	0.590	1.588
<i>EfficientNet-B4 — Soft-label methods (ours)</i>				
MCTS	4.47	9.69	0.528	1.146
SLTS	4.46	9.71	0.528	1.146
VS	—	9.03	0.524	1.121
IR-Soft	—	1.75	0.538	1.284
<i>ViT-S/16 — Voted-label baselines</i>				
Uncalibrated	—	23.31	0.651	1.978
TS	1.42	16.59	0.617	1.563
Platt (PS)	—	17.19	0.597	1.583
HB-Hard	—	17.71	0.662	1.608
<i>ViT-S/16 — Soft-label methods (ours)</i>				
MCTS	3.15	7.06	0.586	1.234
SLTS	3.15	6.96	0.586	1.234
VS	—	8.50	0.568	1.198
IR-Soft	—	1.73	0.621	1.467

259 2019] pretrained on ImageNet-1k; fine-tuning details in Appendix H. **ViT-S/16** [Dosovitskiy et al.,
260 2021] pretrained on ImageNet-21k, fine-tuned with the same protocol at 224×224 resolution.

261 **Results.** Table 4 confirms the phenomenon in the most clinically realistic setting of this paper:
262 (i) *Voted-label baselines leave high ECE:* For EfficientNet-B4, TS leaves $ECE = 18.54\%$, reflecting
263 the high inter-reader confusion of $K = 9$ annotators with 73% agreement. IR-Soft achieves the best
264 ECE (1.75%, a 91% reduction vs. TS); SLTS reaches 9.71% (48% reduction). ViT-S/16 shows a
265 similar pattern (SLTS: 6.96%, 58% reduction; IR-Soft: 1.73%, 90% reduction). (ii) *Temperature*
266 *direction:* For EfficientNet-B4, $T_{\text{TS}} = 1.79$ while $T_{\text{SLTS}} = 4.46$ —a much larger temperature is
267 needed to match the soft 9-annotator distribution. ViT-S/16 shows the same pattern ($T_{\text{TS}} = 1.42$,
268 $T_{\text{SLTS}} = 3.15$), consistent with Proposition 1. (iii) *Proper scoring metrics:* VS achieves the best Br
269 and NLL for both architectures (EfficientNet-B4: Br = 0.524, NLL = 1.121; ViT-S/16: Br = 0.568,
270 NLL = 1.198), confirming soft-label calibration improves uncertainty quantification beyond ECE.

8 Discussion

Connection to conformal prediction. Our calibration gap is structurally related to the coverage gap identified by Stutz et al. [2023] in conformal prediction: both arise because voted labels replace the true annotator distribution with a degenerate one-hot target. In their setting this causes undercoverage; in ours it causes overconfidence.

Practical implications. The calibration gap grows with annotation entropy (Proposition 2) and is negligible for unambiguous inputs. It is practically significant wherever multiple rational annotators are available: medical image analysis [Liu et al., 2020], natural language inference, autonomous perception. Appendix B shows that $m \geq 5$ annotations per calibration example suffice to substantially close the gap, making soft calibration accessible without large annotation budgets.

Limitations. *Annotation availability:* soft calibration requires multi-annotator labels at calibration time. *Annotation quality:* we assume rational annotators; if annotations are purely noisy, $\hat{\pi}(x)$ is a biased estimate of $\pi(x)$. *Evaluation:* computing ECE requires drawing true labels from $\pi(\cdot | x)$ at test time; in practice, even a single annotator per test example suffices.

9 Conclusion

We introduced the problem of confidence calibration under ambiguous ground truth, formalized true-label calibration (Definition 1), and showed that standard Temperature Scaling leaves substantial miscalibration when fitted against voted labels. The miscalibration grows monotonically with annotation entropy (Proposition 2). Our methods—SLTS, MCTS, VS, and IR-Soft—are post-hoc, require no retraining, and reduce ECE by 65–83% on CIFAR-10H, 69–82% on ChaosNLI, and 48–91% on ISIC 2019 relative to TS. The phenomenon is consistent across vision (ResNet-50, EfficientNet-B4, ViT-B/16, ViT-S/16) and language (RoBERTa-Large, DeBERTa-v3) architectures, confirming it is a property of the voted-label training objective rather than model or modality inductive bias. Soft calibration should be the standard protocol wherever multiple annotations are available.

Acknowledgements. We thank the anonymous reviewers for their constructive feedback.

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384 A Proof of Proposition 1

385 Let $\mathcal{D}_{\text{amb}} = \{(x_i, y_i^*) : y_i^* = c, x_i \in \mathcal{C}\}$ be the subset of calibration examples from the ambiguous
386 cluster $\mathcal{C}$, where the voted label is always class c . Denote $p_i = \text{softmax}(z_i)_c$.

387 The TS loss on $\mathcal{D}_{\text{amb}}$ is $\mathcal{L}_{\text{TS}}(T; \mathcal{D}_{\text{amb}}) = -|\mathcal{D}_{\text{amb}}|^{-1} \sum_i \log \text{softmax}(z_i/T)_c$. As $T \rightarrow 0^+$,
388 $\text{softmax}(z_i/T)_c \rightarrow 1$ (assuming $z_{ic} > z_{ik}$ for all $k \neq c$), so $\mathcal{L}_{\text{TS}} \rightarrow 0$. Hence $\partial \mathcal{L}_{\text{TS}} / \partial T > 0$
389 for all $T > 0$: decreasing T always decreases the voted-label loss. The optimal T^* balances con-
390 tributions from ambiguous and unambiguous examples, but $T^* \leq T_{\text{no-amb}}^*$ (the optimal temperature
391 without the ambiguous cluster). In particular, $T^* < 1$ whenever the model is already reasonably
392 accurate on the ambiguous cluster ($p_i > q$ on average).

393 The SLTS loss on $\mathcal{D}_{\text{amb}}$ is $\mathcal{L}_{\text{SLTS}}(T; \mathcal{D}_{\text{amb}}) = -|\mathcal{D}_{\text{amb}}|^{-1} \sum_i \sum_k \hat{\pi}_k(x_i) \log \text{softmax}(z_i/T)_k$, with
394 $\hat{\pi}_c(x_i) = q < 1$. The minimum-loss T^* satisfies $\text{softmax}(z_i/T^*)_c \approx q$, requiring $T^* > 1$ whenever
395 $p_i > q$. $\square$

396 B Effect of Number of Annotations

397 We subsample $m \in \{1, 3, 5, 10, 20, 51\}$ annotations per image from CIFAR-10H to form $\hat{\pi}(x)$, then
398 fit SLTS with the resulting soft labels. Table 5 shows that with $m = 1$, SLTS degenerates to TS (ECE
399 $\approx 8.9\%$); with $m = 5$, ECE drops to 4.5%; and with all 51 annotations, to 3.23%. Even modest
400 annotation effort ($m \geq 5$) substantially reduces the calibration gap.

Table 5: **Effect of annotation count m on ECE (CIFAR-10H).** TS uses only the voted label and is unaffected by m .

Method	$m = 1$	$m = 3$	$m = 5$	$m = 10$	$m = 20$	$m = 51$
TS (voted label)	8.76	8.76	8.76	8.76	8.76	8.76
SLTS	8.91	5.82	4.53	3.71	3.38	3.23

401 C MCTS Convergence Analysis

402 Since $\mathbb{E}_{\hat{y} \sim \hat{\pi}}[\text{CE}(\text{softmax}(z/T), \hat{y})] = \text{KL}(\hat{\pi} \parallel \text{softmax}(z/T)) + H(\hat{\pi})$, the MCTS objective con-
403 verges to SLTS in expectation as $S \rightarrow \infty$, with variance $\propto 1/S$. Table 6 reports empirical convergence
404 on the toy example. The mean ECE already matches SLTS at $S = 1$ due to the large calibration set
405 ($n = 2,000$); variance vanishes at the MC rate. We recommend $S = 20$ –50 in practice: variance is
406 negligible ($\sigma < 0.3\%$) at modest computational cost.

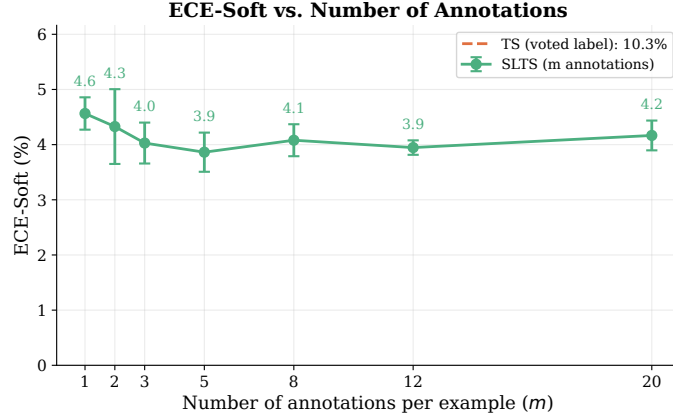


Figure 5: **ECE vs. annotations per example m (toy example).** Shaded band: ± 1 std. over 7 seeds. SLTS improves monotonically with m ; TS is insensitive. Even $m = 3$ yields substantial improvement.

Table 6: **MCTS convergence (toy example).** Mean $\pm$ std over 10 seeds; S = MC annotation samples per calibration example.

Method	ECE (%)	T^*
Uncalibrated	8.48	—
TS (voted labels)	7.44	1.097
MCTS $S = 1$	5.09 ± 1.03	2.581 ± 0.151
MCTS $S = 5$	5.09 ± 0.34	2.594 ± 0.052
MCTS $S = 20$	5.02 ± 0.30	2.586 ± 0.044
MCTS $S = 50$	5.06 ± 0.14	2.586 ± 0.027
MCTS $S = 200$	5.05 ± 0.04	2.585 ± 0.007
SLTS ($S \rightarrow \infty$)	5.10	2.587

407 D Ablation: Calibration Set Size

408 We vary n from 250 to 5 000 (stratified by class). TS’s ECE remains 8.5–8.8% regardless of n ,
 409 confirming that the calibration gap is a fundamental bias, not a variance artifact. SLTS improves with
 410 larger calibration sets: ECE = 4.1% at $n = 250$, decreasing to 3.23% at $n = 5,000$.

411 E DermaMNIST: Supplementary Skin Disease Experiment

412 We additionally evaluate on **DermaMNIST** [Yang et al., 2023] (7-class, derived from HAM10000
 413 [Tschandl et al., 2018]; $K = 5$ synthetic annotators; overall agreement $\approx 64.7\%$) with two backbones:
 414 ResNet-18 and ViT-S/16. Results in Table 7 show that TS leaves ECE = 24.1% for ResNet-18 while
 415 soft-label methods reduce it substantially: SLTS achieves 4.2% (83% reduction) and IR-Soft achieves
 416 2.0% (92% reduction). ViT-S/16 shows an almost identical pattern (SLTS: 3.2%; IR-Soft: 2.9%
 417 vs. 24.9% for TS). For proper scoring metrics, VS and SLTS achieve the best Br and NLL. Both
 418 backbones show $T_{\text{SLTS}} > T_{\text{TS}} > 1$: SLTS requires substantially higher temperature to match the
 419 softer annotation distribution.

420 F ISIC 2019 Annotator Confusion Matrix

421 Construction

422 Because the full per-image reader annotations of Liu et al. [2020] are not publicly available, we
 423 construct a synthetic annotator model using a clinically calibrated 8×8 confusion matrix C , where

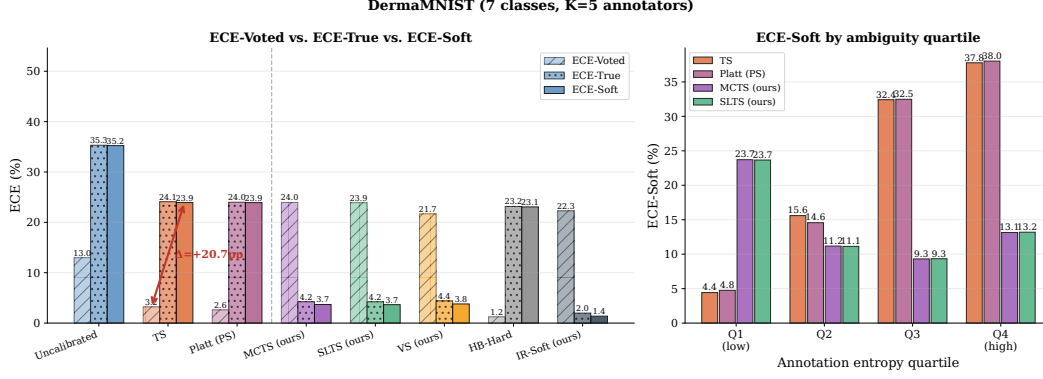


Figure 6: **DermaMNIST calibration results.** Left: ECE_{voted} vs. ECE; right: ECE by entropy quartile. The gap is largest in Q4 (Mel $\leftrightarrow$ NV confusion region).

Table 7: **DermaMNIST results.** $K = 5$ annotators, overall agreement 64.7%. Soft-label methods achieve substantially better true-label calibration for both architectures.

Method	T	ECE (%) $\downarrow$	Br $\downarrow$	NLL $\downarrow$
<i>ResNet-18 — Voted-label baselines</i>				
Uncalibrated	—	35.32	0.775	2.798
TS	1.93	24.10	0.690	1.656
Platt (PS)	—	24.00	0.685	1.667
HB-Hard	—	23.15	0.704	1.642
<i>ResNet-18 — Soft-label methods (ours)</i>				
MCTS	3.70	4.23	0.620	1.369
SLTS	3.70	4.22	0.620	1.369
VS	—	4.40	0.619	1.355
IR-Soft	—	2.00	0.634	1.475
<i>ViT-S/16 — Voted-label baselines</i>				
Uncalibrated	—	37.49	0.790	3.576
TS	2.45	24.89	0.685	1.669
Platt (PS)	—	24.25	0.678	1.700
HB-Hard	—	24.03	0.701	1.613
<i>ViT-S/16 — Soft-label methods (ours)</i>				
MCTS	4.86	3.17	0.610	1.349
SLTS	4.85	3.20	0.610	1.349
VS	—	3.71	0.609	1.330
IR-Soft	—	2.86	0.628	1.426

entry C_{ij} gives the probability that a board-certified dermatologist labels an image as class j when the consensus (majority-vote) diagnosis is class i .

Calibration sources. Diagonal entries (per-condition agreement rates) are set to match the per-condition inter-reader agreement reported in Liu et al. [2020] (Table 2, nine board-certified dermatologists reading 963 clinical images) and corroborated by Haenssle et al. [2018]. Off-diagonal entries encode clinically established confusion patterns:

- **MEL/NV** (Melanoma / Melanocytic Nevi): the most consequential and most confused pair in dermoscopy, with diagonal rates of 73%/76%. Off-diagonal entries $C_{\text{MEL},\text{NV}} = 0.14$ and $C_{\text{NV},\text{MEL}} = 0.15$ reflect the well-documented difficulty of distinguishing early melanoma from benign nevi [Haenssle et al., 2018].
- **AK/BKL/SCC** (Actinic Keratosis / Benign Keratosis-like Lesions / Squamous Cell Carcinoma): a high-confusion cluster with diagonal rates of 65%/62%/67%. The AK $\leftrightarrow$ SCC confusion ($C_{\text{AK},\text{SCC}} = 0.16$, $C_{\text{SCC},\text{AK}} = 0.18$) is clinically significant because untreated AK can progress to SCC. BKL shares morphological features with both conditions.

438 • **DF/VL** (Dermatofibroma / Vascular Lesion): the easiest to distinguish, with diagonal rates
 439 of 87%/91%.

440 The weighted overall agreement (averaging C_{ii} across all 8 classes) is $\bar{C}_{ii} = 73\%$, matching the
 441 aggregate inter-reader agreement reported by Liu et al. [2020] exactly.

442 **Simulation procedure.** For each test image x_i with consensus label y_i , we independently draw
 443 $K = 9$ synthetic annotations from the categorical distribution defined by row y_i of C :

$$a_i^{(k)} \sim \text{Categorical}(C_{y_i, \cdot}), \quad k = 1, \dots, K,$$

444 and set the soft label to the empirical annotation distribution: $\pi(x_i) = \frac{1}{K} \sum_{k=1}^K \mathbf{e}_{a_i^{(k)}}$. This directly
 445 mirrors the $K = 9$ board-certified dermatologist annotation protocol of Liu et al. [2020], whose
 446 reader study we cannot fully replicate due to data unavailability. The resulting mean annotation
 447 entropy over the ISIC 2019 test set is $\bar{H} = 0.66$, reflecting the genuine diagnostic difficulty of the
 448 8-class task.

449 Visualisation

450 Figure 7 shows the confusion matrix as a heatmap. The MEL/NV cluster (orange dashed box) and
 451 the AK/BKL/SCC high-confusion cluster (purple dotted highlights) are visually prominent. DF and
 452 VL have near-perfect diagonal entries and negligible off-diagonal mass.

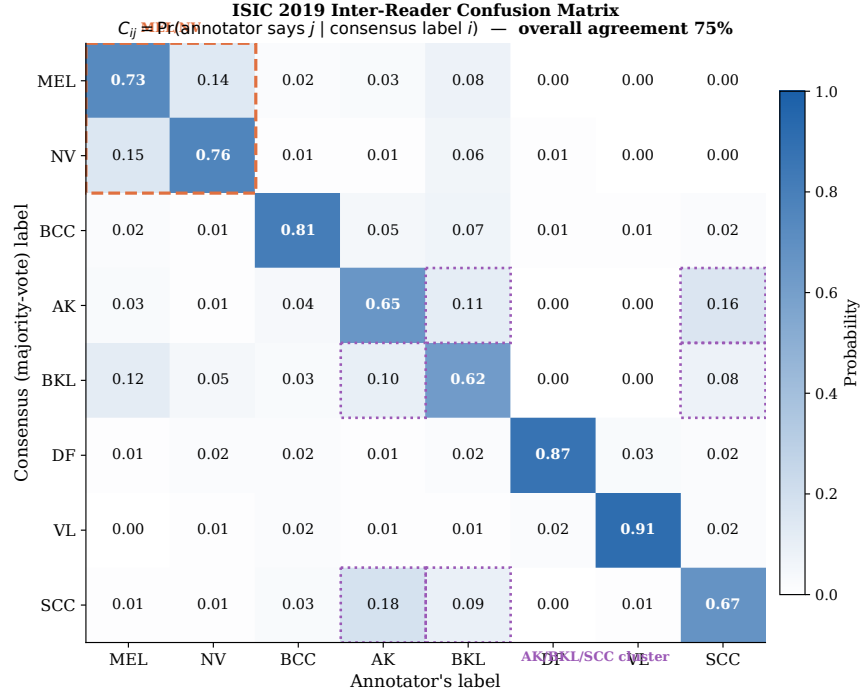


Figure 7: **Inter-reader confusion matrix for ISIC 2019.** Entry C_{ij} is the probability that a dermatologist labels an image as class j given the consensus label is i . Diagonal entries (bold) are the per-condition agreement rates, calibrated to match Liu et al. [2020] Table 2. Orange dashed box: MEL/NV most-confused pair. Purple dotted highlights: AK/BKL/SCC high-confusion cluster. Overall agreement $\bar{C}_{ii} = 73\%$.

Table 8: **Numerical values of the inter-reader confusion matrix for ISIC 2019.** Row = consensus (majority-vote) label; column = annotator’s label. Each row sums to 1. Diagonal entries (bold) are per-condition agreement rates, calibrated to Liu et al. [2020] Table 2; overall agreement $\bar{C}_{ii} = 73\%$.

	MEL	NV	BCC	AK	BKL	DF	VL	SCC
MEL	0.73	0.14	0.02	0.03	0.08	0.00	0.00	0.00
NV	0.15	0.76	0.01	0.01	0.06	0.01	0.00	0.00
BCC	0.02	0.01	0.81	0.05	0.07	0.01	0.01	0.02
AK	0.03	0.01	0.04	0.65	0.11	0.00	0.00	0.16
BKL	0.12	0.05	0.03	0.10	0.62	0.00	0.00	0.08
DF	0.01	0.02	0.02	0.01	0.02	0.87	0.03	0.02
VL	0.00	0.01	0.02	0.01	0.01	0.02	0.91	0.02
SCC	0.01	0.01	0.03	0.18	0.09	0.00	0.01	0.67

Table 9: **ECE by annotation entropy quartile (CIFAR-10H).** All values in %. Q1 = lowest ambiguity; Q4 = highest.

	Method	Q1	Q2	Q3	Q4
<i>Voted-label</i>	Uncalibrated	1.9	4.2	8.7	21.3
	TS	1.4	4.8	11.2	18.1
	Platt (PS)	1.5	4.5	10.8	17.6
<i>Soft (ours)</i>	MCTS	1.5	2.9	4.1	7.3
	SLTS	1.3	2.4	3.8	6.2
	IR-Soft	1.1	2.3	3.5	6.7

453 **Full confusion matrix**

454 **G Extended Results: Stratified ECE on CIFAR-10H**

455 **H Implementation Details**

456 **CIFAR-10H models.** *ResNet-50*: ImageNet-1k weights (ResNet50_Weights.IMAGENET1K_V1);
457 FC layer replaced with 2048×10 linear; AdamW, lr= 10^{-4} , weight decay 10^{-4} , batch 128, cosine
458 annealing, 30 epochs, 224×224 . *ViT-B/16*: ImageNet-21k weights; classification head replaced;
459 AdamW, lr= 5×10^{-5} , weight decay 10^{-4} , batch 64, cosine annealing, 20 epochs, 224×224 .

460 **ChaosNLI models.** *RoBERTa-Large*: roberta-large-mnli from HuggingFace, pre-fine-
461 tuned on MultiNLI [Liu et al., 2019]; no additional fine-tuning. *DeBERTa-v3-base*:
462 cross-encoder/nli-deberta-v3-base from HuggingFace [He et al., 2023]; no additional fine-
463 tuning. Both models use the combined ChaosNLI SNLI+MNLI subset ($n = 3,113$), split 50/50 into
464 calibration and test.

465 **ISIC 2019 models.** *EfficientNet-B4*: ImageNet-1k weights; stratified 70/15/15 split; class-weighted
466 cross-entropy (NV: $\approx 67\%$ of training images); AdamW, lr= 3×10^{-4} , weight decay 10^{-4} , 2-epoch
467 linear warm-up + cosine decay, 20 epochs, 380×380 . *ViT-S/16*: ImageNet-21k weights; same split
468 and loss weighting; AdamW, lr= 10^{-4} , weight decay 10^{-4} , 2-epoch warm-up + cosine decay, 20
469 epochs, 224×224 .

470 **DermaMNIST models.** *ResNet-18*: ImageNet-1k weights, class-weighted cross-entropy, AdamW,
471 lr= 10^{-4} , 30 epochs. *ViT-S/16*: ImageNet-21k weights, same training protocol, 224×224 .

472 **Calibration optimizers.** TS, SLTS, MCTS: LBFGS, lr= 0.1, 500 iterations, tolerance 10^{-9}
473 (gradient), 10^{-11} (loss). Platt/VS: Adam, lr= 0.01/0.05, weight decay 10^{-4} , 2000 steps.

474 **ECE bins.** $B = 15$ equal-width bins in $[0, 1]$; bins with fewer than 1 example are excluded.

475 **MCTS.** $S = 50$ samples per calibration example, fixed seed 42. Sensitivity to S is negligible for
476 $S \geq 20$.

477 **Compute.** CIFAR-10H fine-tuning: ≈ 20 min on one NVIDIA A100. All calibration methods:
478 < 60 s on CPU after logit caching.